

Statistics II: Introduction to Inference

Topic 3

February 13, 2025

MSE and Class of Estimators

- Finding the estimator with the lowest MSE from the entire collection of all estimators is generally infeasible.
- Instead, we focus on the class of estimators, for example, to estimate σ^2 , one class of estimators could be:

$$\textbf{Example-1} : s_{a_n}^2 = \frac{1}{a_n} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad \forall a_n > 0.$$

- Another choice of the class of estimators is the class of **unbiased estimators**,

$$\mathcal{U}_{\psi(\boldsymbol{\theta})} = \{T(\mathbf{X}) : \mathbb{E}_{\boldsymbol{\theta}}[T(\mathbf{X})] = \psi(\boldsymbol{\theta}), \text{ for all } \boldsymbol{\theta}\}.$$

```
1  set.seed(2025)
2  n = 40; sig = sqrt(10);
3  MSE = function(t){
4    niter = 500; temp = rep(0, niter);
5    for (i in 1:niter) {
6      x = rnorm(n, mean = 5, sd = sig)
7      temp[i] = (((1/t)*(sum((x - mean(x))^2)))) - sig^2)^2
8    }
9    return(mean(temp))
10 }
11 t = seq(30, 60, by = 1); mse_val = rep(0, length(t))
12 for (i in 1:length(t)) {
13   mse_val[i] = MSE(t[i])
14 }
15 plot(t, mse_val, type = "l", xlab = "divisor", ylab = "MSE")
```

Illustration

```
1 #the divisor for which MSE is minimum.  
2 t[which(mse_val== min(mse_val))]  
3 > 41
```

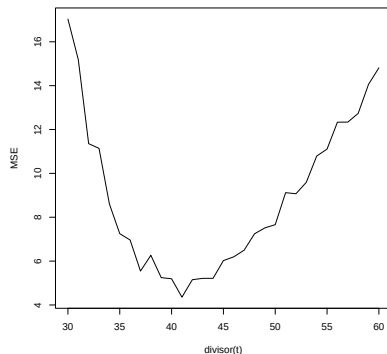


Figure 1: Plotting the MSEs w.r.to different divisors a_n (Example-1)

- Within this class $\mathcal{U}_{\psi(\boldsymbol{\theta})}$, we search for T^* , such that

$$\text{Var}_{\boldsymbol{\theta}}(T^*) \leq \text{Var}_{\boldsymbol{\theta}}(T) \quad \forall T \in \mathcal{U}_{\psi(\boldsymbol{\theta})}.$$

Note that for all $T \in \mathcal{U}_{\psi(\boldsymbol{\theta})}$, the MSE simplifies to:

$$\text{MSE}_{\boldsymbol{\theta}}(T) = \text{Var}_{\boldsymbol{\theta}}(T).$$

- The estimator T^* is called the **Uniformly Minimum Variance Unbiased Estimator (UMVUE)** of $\psi(\boldsymbol{\theta})$.

Cramer Rao Lower Bound (CRLB)

Question: What is the smallest possible variance for unbiased estimators?

Answer: Cramer Rao Lower Bound (CRLB).

Under some **regularity conditions**, for any unbiased estimator $T(\mathbf{X})$ of $\psi(\theta)$,

$$\text{Var}_{\theta}(T) \geq \frac{\{\psi'(\theta)\}^2}{I(\theta)},$$

where $\psi'(\theta) = \frac{\partial}{\partial \theta} \psi(\theta)$ and $I(\theta) = \mathbb{E}_{\theta} \left(\frac{\partial}{\partial \theta} \log f(\mathbf{X}; \theta) \right)^2$.

If an estimator T attains CRLB, it is called an **efficient estimator**. (Here we consider θ is a scalar, not vector.)

- An efficient estimator is always an UMVUE. However, the reverse statement is not always true.
- For the exponential family, the pdf is given by:
$$f(\mathbf{x}; \boldsymbol{\theta}) = \exp \left\{ a(\boldsymbol{\theta}) + b(\mathbf{x}) + \sum_{j=1}^m T_j(\mathbf{x}) c_j(\boldsymbol{\theta}) \right\}; \mathbf{x} \in \mathbb{R}^n$$
- $T(\mathbf{X}) = (T_1(\mathbf{X}), \dots, T_m(\mathbf{X}))$ is jointly complete sufficient.
- Let $T(\mathbf{X})$ be a complete-sufficient statistic, and there exists a function g of $T(\mathbf{X})$ such that, $\mathbb{E}_{\boldsymbol{\theta}}[g(T(\mathbf{X}))] = \psi(\boldsymbol{\theta})$. Then $g(T(\mathbf{X}))$ is the UMVUE of $\psi(\boldsymbol{\theta})$.

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ Binomial (m, p) distribution, where $m \in \mathbb{N}$ is known, but $p \in (0, 1)$ is unknown. Show that $\frac{\bar{X}_n}{m}$ is the UMVUE of p .
2. Consider a random sample of size n from a Geometric distribution with parameter $\theta \in (0, 1)$. Find the UMVUE of θ .
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim}$ Uniform $(0, \theta)$ distribution ($\theta > 0$). Does this distribution belong to the exponential family? Show that $X_{(n)} = \max\{X_1, \dots, X_n\}$ is not an unbiased estimator of θ , but $2\bar{X}_n$ is an unbiased estimator. Which one is better?

4. Do the same using R code as in **Example-1** for
- ① Exponential distribution with rate parameter $\lambda > 0$, and
 - ② Gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$.

Plot the MSEs w.r.t.
different divisors.

5. Do the same using R code as in **Example-1** for a Geometric distribution with parameter $\theta \in (0, 1)$. Plot the MSEs w.r.to different divisors.